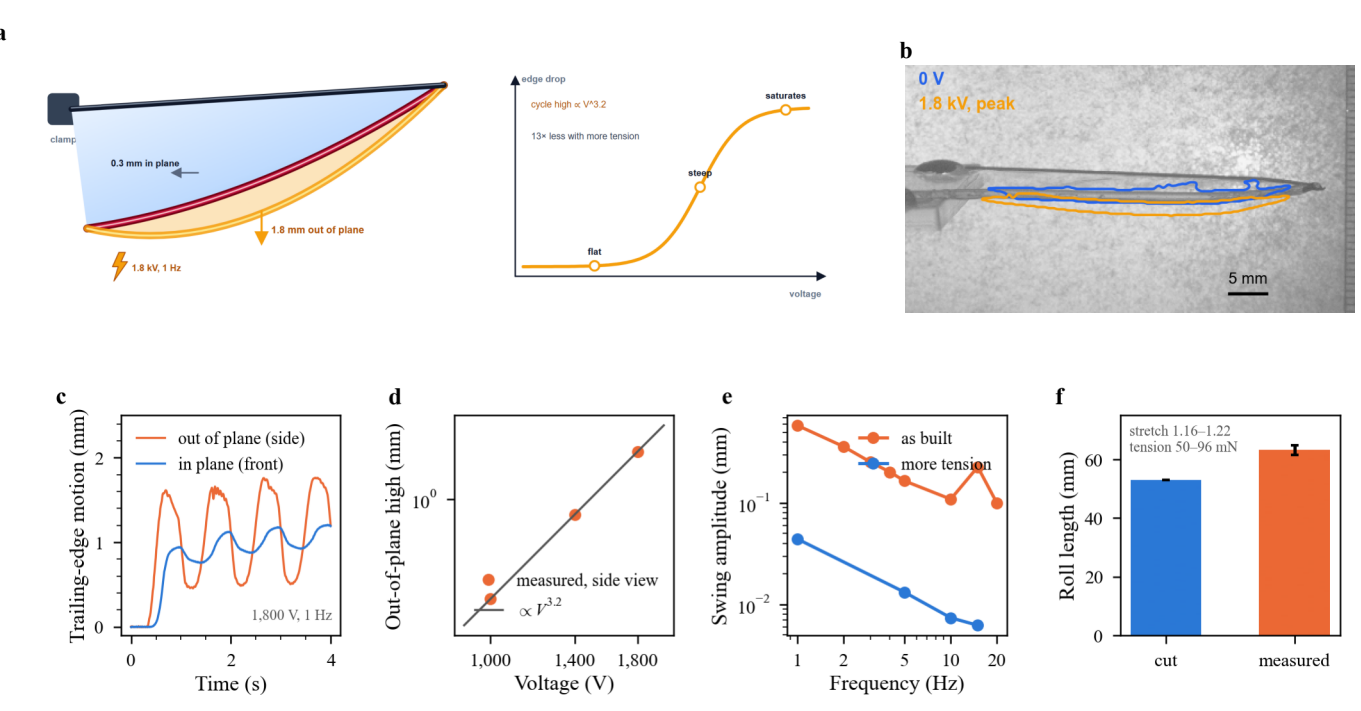

a
clamp
0.3 mm in plane
1.8 mm out of plane
1.8 kV, 1 Hz
edge drop
cycle high ∝ V^3.2
13× less with more tension
flat
steep
saturates
voltage
b
0 V
1.8 kV, peak
5 mm
c
Trailing-edge motion (mm)
out of plane (side)
in plane (front)
1,800 V, 1 Hz
Time (s)
d
Out-of-plane high (mm)
measured, side view
∝ V^3.2
Voltage (V)
e
Swing amplitude (mm)
as built
more tension
Frequency (Hz)
f
Roll length (mm)
stretch 1.16–1.22
tension 50–96 mN
cut
measured